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REGULAR ARTICLE

The Family Socioeconomic Effect on Extra Lessons in Greater China: A Comparison between Shanghai, Taiwan, Hong Kong, and Macao

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Abstract Spawned by the heightened competition in mainstream education, supplementary tutoring outside regular school time has rapidly expanded as a common practice of students seeking to gain a competitive edge in school. It is widely known that students from low socioeconomic status (SES) families lag behind their high-SES peers in academic performance. This study examines whether extra lessons reinforce or weaken the family effect on educational inequality. We use the Programme for International Student Assessment 2009 data to analyze the effect of family SES on the likelihood and intensity of students' extra lesson attendance in four Chinese regions: Shanghai, Taiwan (Chinese Taipei), Hong Kong, and Macao. The research confirms that (1) high-SES students are more likely than low-SES students to receive extra lessons, and (2) among the students who participated, high-SES students also invested more time each week in such lessons. However, the study has also found substantial regional differences.

Keywords Out-of-school-time lessons · Socioeconomic background · Inequality · China

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Introduction

Education plays a central role in deciding individuals' life chances. As a result, competition among students in school is heightened and families tend to mobilize all means possible to help their children gain an edge in school learning. Recent decades have witnessed a worldwide trend of students taking supplementary tutoring classes outside school time to facilitate their learning (Bray 2009; Bray and Lykins 2012). Supplementary tutoring has experienced vigorous growth in many parts of the world, especially in East Asia, where students and families are believed to attach great importance to education because of the cultural influence of the Confucian tradition (Kim and Lee 2010; Liu 2012). Some schools provide extra lessons outside the regular class schedule to their own students, while business also takes advantage of students' need by offering fee-charging tutoring lessons. In Japan, South Korea, and Hong Kong, private tutoring has developed into a billion-dollar commercial industry (Dawson 2010; Jang 2011). Supplementary learning has thrived and grown into a phenomenon that demands social and research attention.

As the popularity of extra learning activities grows, a question arises about educational equality. The extra learning activities are largely spawned by competition in the mainstream schools. And yet there is wide documentation that children's academic achievement in school is heavily influenced by their family backgrounds (Chudgar and Luschei 2009; Coleman 1988; Lareau 2002, 2003). Students from poor families with less-educated parents are disadvantaged in school learning compared with their peers from better-off families (Lareau 2002; Wang 2011). Do extra lessons tend to enlarge or reduce the achievement gap in school caused by students' disparate home conditions? This study aims to explore this question by examining the

patterns of participation in extra lessons among Chinese students. More specifically, this research investigates the effect of family socioeconomic status (SES) on the likelihood and intensity (measured by weekly hours) of extra lessons attendance by students. We employ data from Programme for International Student Assessment (PISA) 2009 to compare the SES effect in four regions in greater China: Shanghai, Taiwan (Chinese Taipei), Hong Kong, and Macao.

In the following section, we first introduce the relevant literature and our hypothesis developed on the basis of the current understanding of extra-school activities in Chinese societies. Then we describe the sampling method, variables, and statistical model. A section on data analysis follows, with a discussion of our findings at the end.

Literature Review

The Global Boom in Extra-School Activities: The Case of Supplementary Tutoring

Although extra-school activities have a long history and a variety of forms, in recent decades one particular form, supplementary tutoring, has experienced a dramatic increase throughout the world. Such tutoring takes many forms. The supplementary lessons can take place in school, at home, in commercial tutoring centers, or elsewhere. From the supply side, the lessons can be provided by schoolteachers to low-performing students or to a whole class in general, by family members or relatives, or by the tutoring industry. Lessons can be fee-charging or free of charge. Overall, supplementary tutoring has silently grown into a striking global phenomenon that is found in many parts of the world, developed economies and low-income countries alike (Bray 2009; Bray and Lykins 2012). A 2007 survey in Japan found 15.9 % of primary students and 65.2 % of secondary students receiving supplementary tutoring in a *juku* (a private tutoring school) (Japan Ministry of Education, Science, Culture, and Sport 2008, 13, cited in Bray 2009). In the Republic of Korea, rates of participation in tutoring have been as high as 87.9 % among elementary school students, 72.5 % among middle school children, and 60.5 % among high school students (Kim and Lee 2010, p. 302). In less-developed Bangladesh, a household survey produced similar statistics: 68.4 % of secondary students in general and more than 80 % of students in grade 10 in particular have used private tutoring services (Nath 2011, cited in Bray and Lykins 2012, p. 4). A 2007 comparative study of former Soviet countries shows that the majority of first-year university students have participated in supplementary tutoring in the last grade of high school. In 2004–2005, the scale of private tutoring ranged from over 80 % in Georgia to over 60 % in Azerbaijan (Silova et al. 2006).[1] The growth of tutoring is also evident in African countries. In 2000, on average 68.3 % of grade 6 children received extra classes in six African countries, compared to 49 % 5 years earlier (Paviot et al. 2008, p. 152). The most dramatic increase was experienced in Malawi, where the participation rate grew from a low 22.1 % in 1995 to 79.7 % in 2000. In Western Europe, North America, and Australia, tutoring remains moderate in scale compared with other regions of the world. However, its expansion is still visible and sometimes is encouraged by the government, as in the case of the United States (Bray 2009).

Extra Lessons in the Chinese Context

In the midst of this global landscape of supplementary tutoring, the greater Chinese region in East Asia has gained special prominence. In mainland China, a survey of 4,472 urban households has found 73.8 % of primary students, 65.5 % of junior secondary students, and 53.5 % of senior secondary students receiving tutoring (Xue and Ding 2009). Similarly, a 2001 survey of 20,000 middle school students in Taipei, China, indicated that 72.9 % of grade 7 children received tutoring for 6.5 h each week on average (Liu 2012). In Hong Kong, a survey in 1996 found 34 % of junior secondary and 40 % of senior secondary students receiving private tutoring (Lee 1996). The figures were more than doubled in 2010, when 82 % of junior secondary and 85 % of senior secondary students participated in private tutoring (Kwo and Bray 2011). The scale of tutoring services in these Chinese societies is estimated to be close to that in other East Asian countries such as Japan, Korea, and Singapore.

The wide popularity of forms of extra learning activities in Chinese communities is believed to be rooted, in part, in the Confucian tradition, which has a long history of examination and tutoring activities (Liu 2012) and emphasizes education and diligence. Such a Confucian legacy is also believed to be one of the reasons why the private tutoring industry has prospered in East Asian countries such as Japan and Korea (Bray 2009; Bray and Lykins 2012). However, the cultural argument encounters significant difficulties in explaining the vigorous growth of private tutoring in dozens of other countries across the world with drastically different cultural and religious traditions (Bray et al. 2013).

In today's Chinese societies, educational qualifications are associated with not only civil servant positions but also

[1] The Georgian number refers to the percentage of respondents in the sample of university students who claimed they received some form of private tutoring.

a wide array of other occupations in the job market. After Chinese societies achieved universal education at the primary and secondary levels, tertiary education credentials have become the key to accessing better-paying white-collar jobs. The mass enrollment in formal schools has intensified competition in university entrance exams as well as in other high-stakes exams at lower levels of education. As Luk (2003, p. 26) remarks, "in the minds of students and parents, all those [educational] qualifications are still the ladders for wealthy and honorable lives…. Most importantly, most parents and teachers still believe that winning in examinations is the destination of education." It is in this context that modern mass education has spawned the mass tutoring services to help students gain a competitive edge in the vehement race for university places and higher positions in the job market.

Implications for Educational Equality

With the vast expansion of supplementary tutoring, parents now have access to an additional avenue to facilitate their children's school learning. The growing dependency of parents and students on supplementary tutoring has resulted in a profound shift, as Kwo and Bray (2011, p. 20) observe, "from the school as the only significant formal center for teaching and learning to a situation in which the shadow sector is a major component and an essential support." The implications are twofold.

On the one hand, the nature of tutoring may influence educational equality. Tutoring is received for either remedial or enhancement purposes. Remedial tutoring has the potential to reduce the inequality in student achievement and offset, to some extent, the impact of family backgrounds on schooling. For low-achieving students, tutoring, if affordable, offers a way to catch up with their peers and improve their grades, if their parents are not able to give them academic support at home. Several Western governments, including those of the United States, Australia, and England, have promoted tutoring programs to remedy the family disadvantages experienced by low-performing students (Bray 2009; Lee 2007). By contrast, enhancement tutoring helps high achievers to advance further and to gain an even greater advantage in the educational competition, and it therefore tends to widen the achievement gap among students. In some East Asian countries, such as Korea and Japan, tutoring is often found to serve high-performing students for the purpose of enhancement.

On the other hand, regardless of the nature of tutoring, access to tutoring services may also have a great impact on educational equality. Schools sometimes offer supplementary tutoring to their own students free of charge, but this represents only a small proportion of tutoring activities. Most tutoring is provided by private or commercial providers against a fee. The pattern that the demand for tutoring is lowest among students from low-SES families and highest among students from high-SES families has been confirmed by studies in African countries (Paviot et al. 2008), in Korea (Kim and Lee 2010), in Japan (Stevenson and Baker 1992), in Turkey (Bray et al. 2013; Tansel and Bircan 2006), and in China (Bray and Kwok 2003; Tsang et al. 2010). Even though remedial tutoring may be helpful for low-achieving students from less-advantaged families, such students are less likely to make use of this assistance. The shadow sector, like the formal school system, is highly influenced by family background.

Not only is tutoring vulnerable to the family effect, it is likely to amplify the impact of family SES. Currently tutoring services are mainly regulated by the market mechanism. Students' access to tutoring and the intensity of their participation are determined by the purchasing power of their families. Thus, the gap between rich and poor families is likely to be exacerbated by the differential access to and use of profit-driven private tutoring. Such a trend has been observed in Japan (Dawson 2010), in Pakistan (Bari 2012), and in India (Bray and Lykins 2012). The trends in Chinese societies have not yet been systematically documented. This study is an effort to fill this gap.

In sum, the present research examines the relationship between family background and students' participation in extra lessons in greater China. Specifically, we analyze the effect of family SES on the likelihood and intensity (in weekly hours of attendance) of students' participation in out-of-school-time lessons. Patterns of participation in extra lessons are compared among four developed Chinese regions, Shanghai, Hong Kong, Taiwan (Chinese Taipei), and Macao, to investigate whether extra lessons tend to reinforce or to reduce inequalities in education. Our hypotheses are the following:

H1 Students from high-SES backgrounds are more likely than students from low-SES backgrounds to participate in out-of-school-time lessons; and

H2 High-SES students spend more time every week on out-of-school-time lessons than do low-SES students.

Methods

We use data from the PISA in 2009 as the main source of information in our analysis. PISA is one of the largest international assessment programs; it is organized by the Organisation for Economic Co-operation and Development (OECD) and carried out on a 3-year cycle. In 2009 it assessed the skills and knowledge of 15-year-old students in 65 nations and territories in tests of language,

mathematics, and science. PISA used a two-stage sampling design to collect information from a representative sample of students in each country or region. In the first stage, schools were selected using systematic probability proportional to size (PPS) sampling. In the second stage, students were randomly sampled with equal probability within each school (OECD 2012). At least 35 students were selected in each school. In 2009, four Chinese regions participated in PISA: Hong Kong, Macao, Shanghai, and Taiwan (Chinese Taipei). We also use data from previous cycles (2006, 2003) to document trends for Hong Kong (2003 and 2006), Macao (2003 and 2006), and Taiwan (2006).[2] A brief description of the four regions is given in Table 1.

The choice of these four regions was guided by the goal of including education systems that serve a majority Chinese population and use Chinese (and its various dialects) as the dominant language. The relative homogeneity of the regions studied in terms of their common Chinese cultural heritage offers an opportunity to examine whether the impact of Confucian culture might differ between societies. Another reason for choosing these four regions is that in all of them, universal education has been achieved at primary and secondary levels and a recently expanded tertiary education sector exists. All the four regions lie toward the affluent end of the education development spectrum, which ensures sufficient investment in their public schools. PISA is currently the only data set that enables this comparison, as it has more Chinese societies among its participants than any similar study.

Variables

Dependent Variables

We obtained information about out-of-school-time lessons from the student questionnaire administered as part of the PISA study to select students. Question 32 (Q32) collected information on students' actual attendance time in such lessons. It asked, "How many hours do you typically spend per week attending out-of-school-time lessons in the following subjects?" Students could choose from (a) Do not attend, (b) Less than 2 h per week, (c) 2–4 h per week, (d) 4–6 h per week, and (e) More than 6 h per week.

The term "out-of-school-time lessons" has a certain ambiguity. Its Chinese translation 課後課程 (*kehou kecheng*) may indicate both after-school lessons offered by the school after the regular school hours and tutoring completely outside the school provided by private or commercial services. There is a possibility that the family effect may be clouded by the school effect. To address this issue, we use a statistical procedure to isolate such effect net of the influence of school characteristics, as will be explained later.

Although some researchers tend to use the PISA questionnaire to refer to "private tutoring" or "supplementary tutoring" (Ünal et al. 2010; Choi et al. 2011), a careful reading of the survey instrument suggests cautionary interpretation of the PISA data. As Bray (2010, p. 7) has pointed out, the PISA questionnaire did not ask about "location, motivation, costs, the nature of learning, or multiple other aspects that many analysts of shadow education would like to have learned about." It is a coarse instrument for the purpose of providing information regarding tutoring. Its main merit is to provide comparable data across educational systems. In fact, the question very explicitly instructed students to include extra lessons that "may be given at your school, at your home or somewhere else" and did not mention at all the degree of supplementation or the privateness of such lessons. In this sense, the PISA measure focused on any extra learning activities outside regular school time. Despite its ambiguity, the PISA measure can still fulfill our purpose of examining the correlation between family SES and students' extra learning opportunities in general, be they private or not. The extra lessons are considered supplementary to the school curriculum because the PISA question specifically asks about information on out-of-school-time lessons in the tested language, mathematics, and sciences, all of which are components of the core curriculum in school systems. However, to avoid confusion, we will use the term "extra lessons/classes" for our dependent variable instead of terms such as "tutoring".

Independent Variables

We use the SES measure provided by the PISA team to test for the family effect on students' education. The SES measure is a composite variable containing three components: parents' highest educational attainment, parents' highest occupational status, and home possession (OECD 2012). It is a continuous index normalized at the international mean of zero and the standard deviation of one. In the analysis, we have compared the patterns of extra lessons participation of students in the top one-third of the SES distribution with those of students in the lowest one-third of the SES distribution in each region.[3]

Other student-level characteristics are also selected as controlling variables, including gender, number of siblings, attitude toward school,[4] whether or not the student repeated

[2] We use only items with the same wording from previous cycles to maintain consistency.

[3] Proper sampling weights were used to create the SES dummy variable.

[4] This index is constructed by the PISA team on a continuous scale. The positive value indicates a better attitude toward school (OECD 2010, p. 295).

Table 1 Region descriptions

	2009 GDP per capita (PPP; US$)	PISA 2009 average score[a]			Population (millions)	Number of students in the sample
		Reading	Mathematics	Science		
Hong Kong	43,153[b]	533	555	549	7.07	4,837
Macao	54,960[c]	487	525	511	0.55	5,952
Shanghai	18,398[d]	556	600	575	23.47	5,115
Taiwan	31,782[e]	495	543	520	23.34	5,831

GDP gross domestic product, *PPP* purchasing power parity

[a] OECD (2010)

[b] International Monetary Fund (2012)

[c] World Bank (2013)

[d] National Bureau of Statistics of China (n.d.)

[e] International Monetary Fund (2012)

a grade, current grade level, and immigration status. These are summarized in Table 5 in the Appendix.

Statistical Models

We have employed three logistic regression models. The first (1) is a binary outcome model to estimate the overall probability of out-of-school-time lessons participation by students of different family SES; and the second (2) is an ordered logistic model to detect the SES effect on the intensity of participation (measured by a categorical variable specifying weekly attendance hours) among students who participated in extra lessons. All the models have controlled for the student-level variables listed above. In Model 2, the subscript i indexed the four categories of answers.

$$\begin{aligned}\text{logit}(\text{Tutoring}) = {} & \beta_0 + \beta_1\text{MidSES} + \beta_2\text{HighSES} \\ & + \beta_3\text{Female} + \beta_4\text{Sibling} \\ & + \beta_5\text{Attitude} + \beta_6\text{Repeat} + \beta_7\text{Grade} \\ & + \beta_8\text{Immig} + \in, \end{aligned} \tag{1}$$

$$\begin{aligned}\text{logit}(\text{Time} \le i) = {} & \beta_{i0} + \beta_{i1}\text{MidSES}_i + \beta_{i2}\text{HighSES}_i \\ & + \beta_{i3}\text{Female}_i + \beta_{i4}\text{Sibling}_i \\ & + \beta_{i5}\text{Attitude}_i + \beta_{i6}\text{Repeat}_i + \beta_{i7}\text{Grade}_i \\ & + \beta_{i8}\text{Immig}_i + \in_i, \quad (i = 1, 2, 3, 4). \end{aligned} \tag{2}$$

It is likely that dependent variable may contain information on extra lessons offered by schools, thus confounding the family effect. To address this problem, we use the fixed-effect model (Ammermueller and Pischke 2009; Hanushek and Woessmann 2010; Ohinata and van Ours 2011) to control explicitly for all of the variation in probability that is caused by school characteristics. This enables us to obtain an estimation net of any school effect. The third model is specified in Model 3. The school fixed effect is represented by school dummy variable a.

$$\begin{aligned}\text{logit}(\text{Time} \le i) = {} & \beta_{is0} + \beta_{is1}\text{MidSES}_{is} + \beta_{is2}\text{HighSES}_{is} \\ & + \beta_{is3}\text{Female}_{is} + \beta_{is4}\text{Sibling}_{is} \\ & + \beta_{is5}\text{Attitude}_{is} + \beta_{is6}\text{Repeat}_{is} \\ & + \beta_{is7}\text{Grade}_i + \beta_{is8}\text{Imming}_{is} \\ & + a_s + \in_{is}, \quad (i = 1, 2, 3, 4). \end{aligned} \tag{3}$$

All the models use proper statistical weights to take account of the complex survey design in PISA (OECD 2009).

Results

Patterns and Scale of Extra Lessons Participation

In all four Chinese societies, out-of-school-time lesson participation rates vary according to subject. For the test language[5] (Fig. 1a), more than half of the surveyed students in Shanghai (54 %), close to half of the students in Taiwan (43 %), and approximately one-third of the students in Hong Kong (30 %) and Macao (32 %) reported taking extra classes in 2009.[6] Compared with the 2006 survey statistics, Taiwan experienced an increase (10 %) while Hong Kong and Macao saw a decline in the participation rate (of 13 and 16 %, respectively).

[5] In Shanghai and Taiwan, the test language is Chinese. In Hong Kong, students can choose English or Chinese, depending on the test given to the students. In Macao, students can choose English, Chinese, or Portuguese, depending on the test given to the students.

[6] The participation rate is calculated by subtracting the proportion of students who chose "Do not attend" from 100 %.

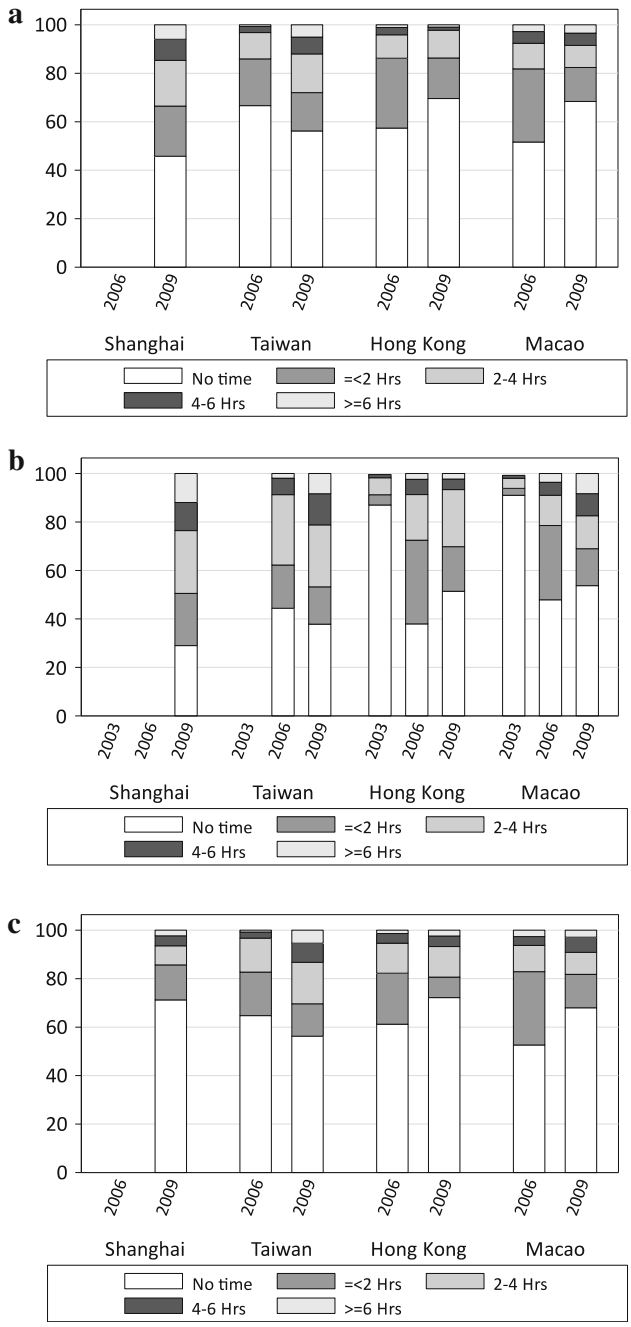

Fig. 1 a. Time spent per week on out-of-school-time test language classes. b. Time spent per week on out-of-school-time math classes. c. Time spent per week on out-of-school-time science classes

Many more students took extra classes in the subject of mathematics, as shown in Fig. 1b. In PISA 2009, Shanghai had the highest participation rate (71 %) in extra mathematics classes, followed by Taiwan (62 %), Hong Kong (49 %), and Macao (46 %). Again, in comparison with PISA 2006, participation in mathematics lessons expanded in Taiwan but shrank in Hong Kong and Macao. However, a contrast with PISA 2003 statistics[7] shows that Hong Kong and Macao had actually gone through a dramatic rise in extra mathematics lessons over the last two cycles of PISA. In 2003 the two cities had only 10 % or so of students taking extra math classes. But the proportions increased rapidly to more than 62 % in Hong Kong and 52 % in Macao in 2006, only to come down slightly to below 50 % in 2009.

Students appeared to be less concerned with the subject of science. Figure 1c shows that more than half of students across all four regions reported no participation at all. The estimated participation rates for extra classes in science are 29, 44, 28, and 32 % in Shanghai, Taiwan, Hong Kong, and Macao, respectively. Again, a comparison of these estimates with those of past years yields a pattern of growth in Taiwan and downturn in Hong Kong and Macao. Taiwan seems to have experienced an expansion in extra classes in all three subjects, while Hong Kong and Macao saw a decline in extra class attendance between 2006 and 2009.

The four regions have great disparities in the intensity of extra-lesson taking measured by weekly hours of attendance. This is clearly exhibited in the relative standings of the four regions in an international comparison. We used the PISA 2009 data to obtain the total time devoted to out-of-school-time classes across all three subjects by each student and calculated the means for each participating country or region. We converted the ordinal values to a continuous scale and then computed a regional average of hours spent on extra lessons each week. For those students who chose "Do not attend," we assigned a value of 0 to indicate nonparticipation. For those who chose "Less than 2 h per week," we assigned 1 as the duration of participation. Similarly, the value "2 to 4 h per week" was coded as 3, "4 to 6 h per week" was coded as 5, and "More than 6 h per week" was coded as 7 h of participation.[8]

The results are plotted in Fig. 2. Three of the four Chinese regions stand above the international average (3.51) in the intensity of extra-lesson taking. Shanghai and Taiwan rank sixth and eighth, respectively, with an average of approximately 6 h of extra lessons per student each week. It is worth noting that 6 h of learning time is equal to an extra school day in many societies. By contrast, Hong Kong and Macao fall markedly behind Shanghai and Taiwan, standing at ranks 17 and 28 with average weekly participation of 4.21 and 3.44 h, respectively.

[7] The survey item first appeared in PISA 2003, which focused on students' math learning. Students were asked to provide out-of-school learning information for math only. Starting with PISA 2006, hours of participation in science and test language classes were also included in the survey.

[8] We acknowledge that this is a rough estimate and do not claim statistical significance.

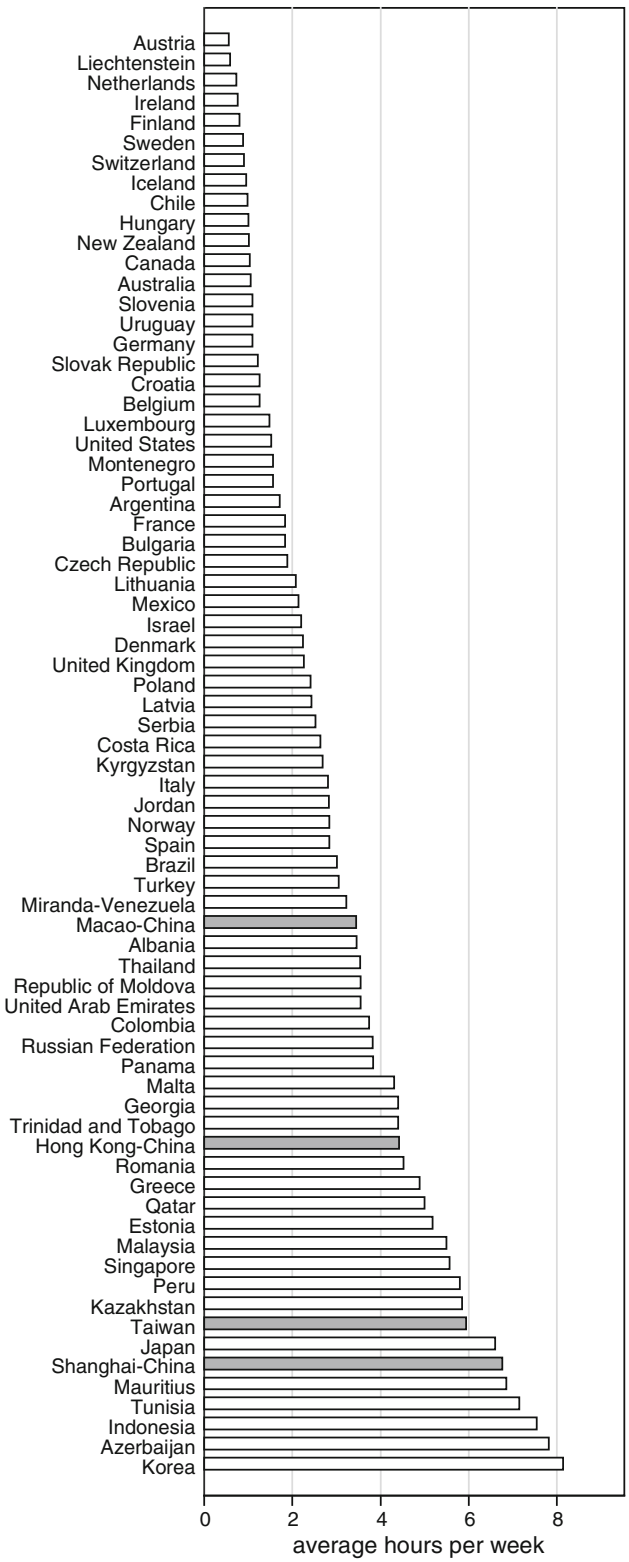

Fig. 2 Estimated hours per week spent on out-of-school-time lessons based on PISA 2009, all subjects combined

Following this method, we estimated the time students spent on extra lessons in each subject in the four regions. In 2009 Shanghai students are estimated to have spent an average of 1.63, 2.41, and 0.75 h per week on the test language, mathematics, and science classes, respectively, versus 1.34, 2.15, and 1.42 h in Taiwan; 0.64, 1.27, and 0.85 h in Hong Kong; and 0.91, 1.60, and 0.93 h in Macao. In comparison with past years, Taiwanese students exhibited a trend of increased supplementary class time. The percentage of students choosing "2 to 4 h per week" and "4 to 6 h per week" for language lessons increased from 11 and 2.5 % in 2006 to 16 and 7 % in 2009. Similarly, the statistics for mathematics lessons show major increases in the categories of "4 to 6 h per week" and "More than 6 h per week," from 6.8 and 2.0 % in 2006 to 13 and 8.4 % in 2009. A similar trend can be observed in science (see Fig. 1c). By contrast, Hong Kong and Macao experienced a drop in extra classes participation between 2006 and 2009. The shrinkage took place mainly among students with minimal attendance in supplementary classes (less than 2 h per week), while the attendance of more dedicated students (those spending more than 2 h in extra classes each week) remained relatively stable, with slight growth or decline (see Fig. 1a–c).

Family SES Effects on Extra Lessons

The first model looks at the effect of family SES on the probability of students participating in classes outside school time, controlling for individual-level characteristics. Students who did not attend any such classes were coded as 0. The rest of the students were coded as 1, regardless of the actual amount of time they spent on supplementary classes. The results are reported in Table 2 in the log-odds metric. The SES effects are pronounced and consistent in mathematics and science across all four societies. Students in the top one-third SES category have a higher probability of attending extra classes in mathematics and science than do students of middle SES. Both groups are more likely than low-SES students to receive extra lessons. Compared to low-SES students, the odds of high-SES students participating in extra math lessons in the four Chinese societies are 1.56–2.1 times higher. However, no association between SES and participation in extra language classes has been found in these Chinese societies except for Hong Kong. High-SES students in Hong Kong are also more likely than low-SES students to participate in lessons in language outside of school time, but there is no statistically significant difference between the middle-SES and low-SES groups.

In addition to the SES effect, students with a positive attitude toward school and students in lower grades of schooling are also more likely to attend extra classes in all subjects except for science in Hong Kong. The gender effect varies in different societies and in language and mathematics. However, female students consistently have a

Table 2 Logistic regression estimates of the odds of a student's participation in supplementary tutoring by subject and region

Variable	Subject/region											
	Language				Mathematics				Science			
	Shanghai	Taiwan	Hong Kong	Macao	Shanghai	Taiwan	Hong Kong	Macao	Shanghai	Taiwan	Hong Kong	Macao
SES dummy[a]												
Middle one-third	0.95	1.07	1.15	1.11	1.22*	1.46***	1.28***	1.01	1.03	1.51***	1.11	1.22**
	(0.08)	(0.08)	(0.11)	(0.08)	(0.10)	(0.11)	(0.07)	(0.07)	(0.09)	(0.13)	(0.09)	(0.09)
Top one-third	1.04	1.04	1.52***	1.11	1.56***	2.10***	1.77***	1.21**	1.19+	1.82***	1.73***	1.30**
	(0.08)	(0.10)	(0.16)	(0.08)	(0.14)	(0.20)	(0.15)	(0.08)	(0.12)	(0.19)	(0.16)	(0.10)
Female	0.99	1.04	0.76***	1.28***	1.21**	0.95	1.09	1.26***	0.71***	0.80**	0.71***	0.84**
	(0.06)	(0.07)	(0.06)	(0.08)	(0.08)	(0.08)	(0.07)	(0.07)	(0.05)	(0.06)	(0.05)	(0.05)
Has siblings	0.95	1.23+	0.96	0.87	0.85+	1.17	0.97	0.97	1.04	1.21	1.00	0.97
	(0.09)	(0.13)	(0.08)	(0.08)	(0.08)	(0.14)	(0.07)	(0.08)	(0.11)	(0.14)	(0.09)	(0.08)
Attitude toward school	1.31***	1.14***	1.09+	1.13**	1.22***	1.04	1.10+	1.10*	1.28***	1.06	1.10*	1.05
	(0.05)	(0.04)	(0.05)	(0.04)	(0.05)	(0.04)	(0.05)	(0.04)	(0.05)	(0.04)	(0.05)	(0.04)
Ever repeated grade	0.73+	0.84	1.12	0.99	0.72*	0.45*	0.87	1.01	1.23	0.38**	0.75*	1.02
	(0.13)	(0.36)	(0.12)	(0.09)	(0.12)	(0.16)	(0.09)	(0.09)	(0.19)	(0.14)	(0.10)	(0.10)
Grade	0.65***	0.40***	1.12	0.76***	0.58***	0.39***	1.16+	0.87*	1.17	0.16***	1.40***	0.95
	(0.05)	(0.04)	(0.11)	(0.05)	(0.05)	(0.04)	(0.09)	(0.05)	(0.13)	(0.01)	(0.12)	(0.05)
Immigration status[b]												
Second generation	0.24	1.17	0.86+	1.16*	0.00	1.44	0.84*	1.12	4.38	0.85	0.87	0.98
	(0.29)	(0.67)	(0.07)	(0.08)	0.00	(1.02)	(0.07)	(0.09)	(6.63)	(0.56)	(0.08)	(0.08)
First generation	1.32	0.20	0.91	1.16	5.26	0.45	0.77*	1.04	2.80*	0.27**	0.70**	0.97
	(0.73)	(0.31)	(0.11)	(0.11)	(6.69)	(0.30)	(0.08)	(0.10)	(1.37)	(0.11)	(0.09)	(0.10)
N	4,116	4,524	4,444	4,371	4,156	4,583	4,436	4,458	4,131	4,513	4,417	4,336

Standard errors in parentheses

[a] Bottom one-third as reference group

[b] Native students as reference group

+ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 3 Ordered logistic regression on the intensity of tutoring participation

Variable	Subject/region											
	Language				Mathematics				Science			
	Shanghai	Taiwan	Hong Kong	Macao	Shanghai	Taiwan	Hong Kong	Macao	Shanghai	Taiwan	Hong Kong	Macao
SES dummy[a]												
Middle one-third	1.05	1.18^{+}	0.87	1.06	1.21*	1.33**	1.24^{+}	1.35**	1.47*	1.16	1.15	1.02
	−0.09	−0.11	−0.13	−0.13	−0.09	−0.12	−0.15	−0.13	−0.25	−0.16	−0.17	−0.13
Top one-third	1.21^{+}	1.68***	1.09	1.12	1.34***	1.80***	1.54***	1.41***	1.75***	1.78***	1.47**	1.06
	−0.12	−0.16	−0.18	−0.14	−0.11	−0.16	−0.17	−0.14	−0.27	−0.22	−0.20	−0.12
Female	1.08	1.09	0.93	1.13	1.18*	0.97	1.07	1.00	0.71**	0.91	1.06	0.89
	−0.09	−0.09	−0.11	−0.11	−0.09	−0.08	−0.08	−0.08	−0.09	−0.08	−0.14	−0.08
Has siblings	1.13	1.17	1.01	1.10	1.02	1.04	0.85	1.00	0.94	1.29^{+}	1.12	0.94
	−0.13	−0.18	−0.13	−0.14	−0.08	−0.14	−0.09	−0.10	−0.14	−0.17	−0.11	−0.11
Attitude toward school	1.03	1.15*	1.07	1.15^{+}	1.04	0.96	1.12^{+}	1.18**	0.90	0.99	1.15*	1.09
	−0.05	−0.06	−0.08	−0.09	−0.05	−0.04	−0.07	−0.07	−0.07	−0.06	−0.08	−0.07
Ever repeated grade	0.96	0.62	0.89	1.28	0.78	0.41^{+}	0.96	1.09	0.99	0.37^{+}	1.50^{+}	0.87
	−0.16	−0.26	−0.23	−0.19	−0.12	−0.21	−0.18	−0.14	−0.25	−0.19	−0.34	−0.13
Grade	0.73***	0.53***	0.79	1.02	0.94	0.60***	0.96	1.12^{+}	1.01	0.29***	1.57***	1.00
	−0.06	−0.05	−0.11	−0.09	−0.06	−0.05	−0.09	−0.07	−0.13	−0.03	−0.20	−0.09
Immigration status[b]												
Second generation	1.02	0.68	0.82^{+}	0.89	3.51	1.53	0.85	0.91	0.00	5.56^{+}	0.73^{+}	0.89
	−0.13	−0.97	−0.1	−0.11	−3.07	−0.69	−0.09	−0.09	0.00	−5.29	−0.13	−0.10
First generation	1.42	0.00	0.78	1.48*	1.66	2.24	0.73*	0.98	2.03	2.38	0.77	0.78
	−0.62	0.00	−0.16	−0.22	−0.85	−1.71	−0.11	−0.13	−1.12	−1.28	−0.16	−0.15
N	2,253	1,938	1,354	1,376	2,977	2,840	2,161	2,067	1,207	1,922	1,232	1,390

Standard errors in parentheses

[a] Bottom one-third as reference group

[b] Native students as reference group

$^{+}$ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 4 Ordered logistic regression on the intensity of tutoring participation with school fixed effects

Variable	Subject/region											
	Language				Mathematics				Science			
	Shanghai	Taiwan	Hong Kong	Macao	Shanghai	Taiwan	Hong Kong	Macao	Shanghai	Taiwan	Hong Kong	Macao
SES dummy[a]												
Middle one-third	1.10	1.21	0.80	1.02	1.22*	1.34**	1.10	1.25*	1.61**	1.14	1.07	0.98
	−0.12	−0.15	−0.14	−0.13	−0.12	−0.15	−0.14	−0.13	−0.27	−0.16	−0.19	−0.13
Top one-third	1.31*	1.77***	0.84	1.07	1.32**	1.67***	1.29+	1.14	1.85**	1.73***	1.15	0.98
	−0.17	−0.23	−0.16	−0.15	−0.14	−0.19	−0.18	−0.13	−0.36	−0.25	−0.21	−0.14
Female	1.10	1.02	0.90	1.06	1.14+	1.03	1.01	0.96	0.71**	0.98	1.25	0.91
	−0.09	−0.1	−0.13	−0.13	−0.08	−0.09	−0.11	−0.09	−0.09	−0.1	−0.18	−0.11
Has siblings	1.22+	1.29	0.99	1.10	1.02	1.05	0.85	0.97	0.96	1.29	1.20	0.93
	−0.15	−0.23	−0.15	−0.15	−0.11	−0.14	−0.09	−0.1	−0.18	−0.23	−0.18	−0.12
Attitude toward school	1.05	1.17**	1.06	1.15+	1.09+	1.01	1.15*	1.20**	0.84*	1.04	1.23*	1.09
	−0.05	−0.07	−0.10	−0.09	−0.05	−0.05	−0.07	−0.07	−0.06	−0.06	−0.11	−0.08
Ever repeated grade	1.01	0.68	0.87	1.33+	0.78	0.51	1.06	1.12	1.07	0.50	1.42	0.83
	−0.19	−0.33	−0.23	−0.20	−0.13	−0.31	−0.19	−0.14	−0.34	−0.33	−0.42	−0.13
Grade	0.76*	0.60*	0.81	1.00	1.01	0.83	1.03	1.02	0.81	0.36***	1.65**	1.00
	−0.10	−0.13	−0.12	−0.09	−0.12	−0.14	−0.10	−0.07	−0.17	−0.08	−0.25	−0.10
Immigration status[b]												
Second generation	0.76	1.18	0.79	0.93	2.97	1.34	0.86	0.91	0.00	3.31	0.71*	0.86
	−0.36	−1.38	−0.13	−0.12	−3.31	−0.62	−0.1	−0.09	0.00	−3.34	−0.12	−0.10
First generation	2.33*	0.00	0.70	1.67**	2.37	1.75	0.78	1.07	1.29	0.82	0.67	0.77
	−0.9	0.00	−0.16	−0.29	−1.30	−1.39	−0.13	−0.15	−0.68	−0.70	−0.18	−0.13
N	2,253	1,938	1,354	1,376	2,977	2,840	2,161	2,067	1,207	1,922	1,232	1,390

Standard errors in parentheses

[a] Bottom one-third as reference group

[b] Native students as reference group

+ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

lower probability of taking extra classes in science. Interestingly, students in higher school grades are less likely to participate in extra lessons, contrary to the received wisdom according to which students in higher grades would be driven to more extra lessons because of the increasing pressure of examinations.

Then we modeled the SES effect on the actual time invested every week in extra lessons in each subject. We excluded the students who reported not attending any out-of-school-time lessons and restricted our analysis to the students who did take extra classes. As there are only four values for the dependent variable (less than 2 h, 2–4 h, 4–6 h, and more than 6 h), we treat it as an ordinal rather than a continuous variable. The estimates are summarized in Table 3, again in the log-odds metric. Family SES is positively correlated with the number of hours spent on mathematics and science classes in Shanghai, Taiwan, and Hong Kong, but not so much in Macao. For instance, in Shanghai, the estimates suggest that for high-SES students, other things being equal, the odds of spending more time in extra mathematics lessons are 1.34 times higher than the odds for low-SES students. Similarly, the odds of high-SES students participating in more extra science classes are 1.75 times higher than the odds of low-SES students doing the same. A similar effect persists in the other three societies. Middle-SES students also tend to invest more time in mathematics classes outside school time compared with low-SES students, but the correlation does not hold for science classes. Interestingly, the SES effect is not strong in the subject field of language. Except in the case of Taiwan, the model identifies no statistically significant correlation between SES background and weekly hours of extra lessons participation in language subject. Gender, number of siblings, and grade repetition generally have little systematic impact on the intensity of extra classes participation. Attitude toward school is positively associated with time spent on out-of-school classes, but the effect does not hold consistently across all four regions. As in Model 2, in Model 3 a student's grade level is negatively correlated with extra classes participation. Students in higher grades tend to devote less time to supplementary classes.

In Model 3 we introduced a series of school fixed effects to control for the variation caused by unobserved or omitted school factors. In essence, we forced the comparison among students within schools, not across schools, to make the SES estimates more accurate. The results are summarized in Table 4.

Controlling for school fixed effects barely changes the magnitudes of the SES coefficients. This suggests that the results from Model 2 are in general robust in these regions. After the school context is accounted for, regional differences become more conspicuous in the SES effect. The significance of the SES effect diminishes in Hong Kong and Macao but not in Shanghai or Taiwan. High-SES students tend to have a higher likelihood of taking after-school classes in all three subjects, including language, in Shanghai and Taiwan, whereas in Hong Kong and Macao no such effect is found in any subject. These findings may indicate that family background plays a more important role in determining the intensity of extra classes in Shanghai and Taiwan than it does in Hong Kong and Macao. We do not observe any clear gender, sibling, or grade repetition effects.

Conclusions

This study investigates the patterns of students' attendance in extra lessons in four Chinese regions. Our findings reveal medium to high participation rates among 15-year-olds in classes outside school hours, ranging from approximately 30–70 % depending on subject. This means that a significant proportion of 15-year-old students in these Chinese societies were attending extra classes in 2009. Shanghai and Taiwan had higher participation rates in general than did Hong Kong or Macao in 2009, but the latter two societies had experienced significant expansion in extra classes attendance since 2003. The intensity of participation in such classes is higher in these societies than in the majority of countries and regions in PISA. Students in Shanghai and Taiwan spent on average around 6 h per week on extra lessons, while for Hong Kong and Macao the number was around 4 h. Across all four societies, mathematics appears to be the most popular subject for extra classes in terms of both the scale and the intensity of student participation.

We are particularly interested in the influence of family background on the likelihood and intensity of taking extra classes. The analyses have confirmed our first hypothesis, namely, that family SES is positively associated with the probability of extra lessons participation in general. In all four societies, high-SES students are more likely than low-SES students to take out-of-school-time classes in mathematics and science, though not in language. Female students have a lower probability than male students of taking extra classes in science, but not necessarily in language or mathematics. Positive attitudes toward school increase students' likelihood of receiving extra lessons. In addition, the higher the grade a student is in, the lower the likelihood of extra lesson participation.

Hypothesis 2 is confirmed in the cases of Shanghai and Taiwan, but not in those of Hong Kong or Macao. In Shanghai and Taiwan, students with high family SES tend to receive more after-school classes than do students from low-SES families in all three subjects. The results are robust even when school fixed effects are controlled for.

The picture in Hong Kong is less clear. The SES effect is statistically significant in mathematics and science, but it disappears once the school fixed effects are accounted for. In Macao, family SES affects the intensity of extra class taking only in mathematics before controlling for the school fixed effects and has no effect at all afterward.

In sum, the PISA data indicate that students in the four Chinese societies have medium to high rates of participation in and intensity of extra classes. Like mainstream schooling, out-of-school-time lessons are also subject to the influence of students' family backgrounds, thus holding high potential to reproduce or enlarge the academic achievement gaps between high-SES and low-SES students in these Chinese societies. However, this study has found substantial regional disparities. Extra class participation is a more common phenomenon among 15-year-old students in Shanghai and Taiwan than among those in Hong Kong or Macao. The family effect is more conspicuous in Shanghai and Taiwan than in Hong Kong or Macao in influencing students' behaviors in extra-lesson taking.

Discussion

For people concerned with educational equality, this study has generated worrisome results. The function of mainstream schooling in social reproduction has been widely documented (Bourdieu and Passeron 1990). Working-class students are disadvantaged in both economic and cultural terms compared with their middle- and upper-class counterparts (Lareau 2003). The booming supplementary learning industry could be an opportunity to help the low achievers catch up. Unfortunately, while the PISA data did not directly speak to the issue of tutoring, they showed that extra learning activities are by no means immune to the influence of students' family backgrounds at least in these urban Chinese societies. Students in better-off families with better-educated parents enjoy more access to extra classes and devote more time to out-of-school learning. The pattern of participation has the tendency to reinforce rather than reduce the already existing educational gaps caused by unequal family resources.

Interestingly, the study has found substantial regional differences in these Chinese societies. Hong Kong and Macao lag far behind Shanghai and Taiwan in both participation rates and the intensity of extra classes, at least in 2009. This phenomenon calls for attention to the different social milieus in which education takes place. For instance, Macao, the smallest region in our sample with a population of about 555,700 in 2011 (World Bank 2013), has the lowest estimated rate of participation in out-of-school-time classes. We speculate that the pattern in Macao is due to a relatively lower level of competition in schools and in the broader society. Macao has an exceptionally low unemployment rate, about 1.9 % (Government of Macao Special Administrative Region Statistics and Census Service 2013), and a relatively equal income distribution. There is no centralized standardized test to determine admission to high school. Therefore, students and their families may feel less compelled to seek supplementary learning opportunities.

By contrast, Shanghai and Hong Kong are both highly stratified in terms of income distribution, and competition for university admission is heightened in both societies. Yet the two cities exhibit surprisingly different patterns in extra lessons participation. This finding is also counterintuitive since Hong Kong, firmly committed to the *laissez-faire* principle, might be expected to show a stronger family effect in extra lessons consumption. We speculate that the differences between Shanghai and Hong Kong may be rooted in differences in school structure and testing arrangements. The 15-year-olds in the PISA survey are attending junior secondary grades. In Shanghai, a high proportion of junior secondary schools are separate from high schools. Centralized examinations are administered every year to allocate junior secondary graduates to high schools. In Hong Kong, by contrast, most secondary schools have both junior and senior grades, and no territory-wide examinations are needed to reallocate students at their promotion to senior grades. Therefore, the junior secondary students in Shanghai may experience higher examination pressures compared with their counterparts in Hong Kong, thus driving up participation in and intensity of extra learning in Shanghai.[9] Whether such different patterns persist at the high school level remains to be investigated in future research.

The regional disparities also cast doubt on the cultural argument based on Confucianism in explaining out-of-school-time lesson practices among Chinese learners. Historically in China, extended families formed the basic units for transmitting social and economic resources into education (Campbell and Lee 2003, 2008, 2011; Hao and Clark 2012; Lee 2012). If the strong family effect in Shanghai and Taiwan were a modern reiteration of the old Confucian tradition, would the weak family effect in Hong Kong and Macao indicate a loss or withering of this tradition? In any case, these regional differences remind us that an abstract cultural function of the Confucian heritage cannot be assumed and may be insufficient to illuminate the diverse patterns displayed by various Chinese societies. Empirical research is needed to pinpoint whether and how Confucian values are enacted under different social and political conditions and in different educational institutions.

[9] A recent study in Hong Kong observes such a hike in participation between grade 9 and grade 12 students (Zhan et al. 2013).

At the time of our writing, the literature on extra lessons in Chinese societies is still developing. While there is more background literature on the issue for some regions (Bray and Kwok 2003; Bray et al. 2014; Kwo and Bray 2011; Liu 2012), the systemic factors that contribute to the prevalence of such practices have yet to be identified conclusively. Our study provides new evidence from a comparative perspective to guide the future development of the field and to identify policy levers that might affect the demand for extra lessons.

Certain caveats in our analysis are worth discussing. The PISA student survey does not collect precise information on the number of hours of extra classes taken by students. Students were simply asked to give their best estimate of the time they spent, and only four time categories were provided in Q32 for students to choose from. Therefore, we were able to test the SES effect only in odds ratio metrics and had to rely on rough estimates of general time expenditure on extra lessons. We tried our best to separate out school-related effects on taking extra lessons, but the statistics are only as good as the survey instrument. Instruments that define more precisely the types of supplementary programs could be much more helpful for researchers to learn about the impact and implications of such programs.

This study focuses on the relationship between family background and extra lesson participation instead of examining the effect of such participation on achievement. Because of the complex nature of student learning, social class advantage does not necessarily lead to advantages in educational outcomes (Lareau 2003). In fact, a recent study using a rigorous statistical design shows heterogeneous outcomes among tutoring participants in China (Zhang 2013). More studies are needed on this topic.

We also acknowledge that our findings may have limited generalizability outside the four urban regions included in this study. Patterns revealed in this research may not hold true for less-developed communities in greater China. Fortunately more Chinese regions are lining up for future PISA studies. For a country as large and diverse as China, more cross-regional comparisons are needed to investigate fully the scope and patterns of family involvement in supplementary learning activities.

Appendix

See Table 5.

Table 5 Description of independent variables

Variable	Region			
	Shanghai	Taiwan	Hong Kong	Macao
Proportion				
Female (1 = yes)	50.45 %	49.49 %	47.08 %	49.39 %
Has siblings (1 = yes)	15.87 %	89.61 %	77.95 %	81.49 %
Ever repeated grade (1 = yes)	7.47 %	0.02 %	15.56 %	43.59 %
Grade				
3 grades below modal grade	1.00 %	0.00 %	1.70 %	6.71 %
2 grades below modal grade	4.06 %	0.10 %	7.18 %	19.17 %
1 grade below model grade	37.44 %	34.42 %	25.16 %	34.91 %
Modal grade	57.10 %	65.46 %	65.90 %	38.69 %
1 grade above modal grade	0.41 %	0.02 %	0.06 %	0.52 %
2 grades above modal grade	0.02 %	0.00 %	0.00 %	0.00 %
Immigration status				
Native	99.46 %	99.59 %	60.61 %	29.59 %
Second generation	0.08 %	0.25 %	23.91 %	54.91 %
First generation	0.47 %	0.02 %	15.48 %	15.49 %
Mean				
Attitude toward school	−0.4	−0.34	−0.49	−0.45
Sample size	5,115	5,831	4,837	5,952

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